

Yiqi Liu

CONTACT INFORMATION	404 Uris Hall Ithaca, NY 14850	y13467@cornell.edu yiqi-liu.github.io
INTERESTS	Econometrics : causal inference, partial identification, machine learning/nonparametric methods	
EDUCATION	Cornell University , Ithaca, NY Ph.D. in Economics	Expected 2026
	Tulane University , New Orleans, LA B.S. in Mathematics and Economics with minor in Studio Art, <i>summa cum laude</i>	2020
WORKING PAPERS	Synthetic Parallel Trends (Job Market Paper) [arXiv] <i>Abstract</i> : Popular empirical strategies for policy evaluation in the panel data literature—including difference-in-differences (DID), synthetic control (SC) methods, and their variants—rely on key identifying assumptions that can be expressed through a specific choice of weights ω relating pre-treatment trends to the counterfactual outcome. While each choice of ω may be defensible in empirical contexts that motivate a particular method, it relies on fundamentally untestable and often fragile assumptions. I develop an identification framework that allows for all weights satisfying a <i>Synthetic Parallel Trends</i> assumption : the treated unit’s trend is parallel to a weighted combination of control units’ trends for a general class of weights. The framework nests these existing methods as special cases and is by construction robust to violations of their respective assumptions. I construct a valid confidence set for the identified set of the treatment effect, which admits a linear programming representation with estimated coefficients and nuisance parameters that are profiled out. In simulations where the assumptions underlying DID or SC-based methods are violated, the proposed confidence set remains robust and attains nominal coverage, while existing methods suffer severe undercoverage. Inference for an Algorithmic Fairness-Accuracy Frontier (with Francesca Molinari) [arXiv] - Extended abstract in <i>Proceedings of the 25th ACM Conference on Economics and Computation</i> Using Forests in Multivariate Regression Discontinuity Designs (with Alice Y. Qi) [arXiv]	
WORK IN PROGRESS	Partial Identification of Algorithmic Frontiers with Selective Labels (with Francesca Molinari and Amilcar Velez)	
PROFESSIONAL ACTIVITIES	Conference/Workshop Presentations : - 2025 : Econometric Society World Congress, Stanford Data-Driven Seminar - 2024 : The 25th ACM Conference on Economics and Computation, Bravo/SNSF Workshop on Using Data to Make Decisions at Brown University, ESIF Economics and AI+ML Meeting <u>Journal Referee</u> : <i>Journal of Political Economy</i> , <i>Review of Economic Studies</i> <u>Conference Reviewer</u> : <i>2025 ACM Conference on Fairness, Accountability, and Transparency (FAccT)</i>	
FELLOWSHIPS & AWARDS	Ernest Liu '64, Ta-Chung and Ya-Chao Liu Memorial Fellowship, Cornell Tapan Mitra Memorial Prize for Outstanding Third Year Paper, Cornell The Louis Walinsky Fund in Economics Outstanding Teaching Award, Cornell Sage Fellowship, Cornell Research Award in Economics, Tulane Phi Beta Kappa, Tulane Presidential Scholarship Award, Tulane	Spring 2024 2023 2022 2020-2025 2020 2020 2016-2020

RESEARCH & TEACHING EXPERIENCE	Research Assistant for Levon Barseghyan and Francesca Molinari		Fall 2022-Fall 2023
	Research Assistant for Francesca Molinari		Summer 2022, 2023, 2024
	Teaching Assistant for Douglas McKee		Spring 2022
	- ECON 3120 : Applied Econometrics, undergraduate (grading and office hours only)		
	Teaching Assistant for Suraj Malladi		Fall 2021
	- ECON 6170 : Mathematical Economics, first-year Ph.D. core (evaluation : 4.93/5)		
ADVISING COMMITTEE	Prof. Francesca Molinari (Chair)	Prof. Levon Barseghyan	Prof. José Luis Montiel Olea
	Department of Economics Cornell University fm72@cornell.edu	Department of Economics Cornell University lb247@cornell.edu	Department of Economics Cornell University jlo67@cornell.edu